

Measuring Thematic Drift in Computational Linguistics

Alignment of TACL Articles with Journal Aims & Scope · 2015–2025

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Why Study Thematic Drift?

The NLP Landscape Has Changed

Rule-based NLP → Deep Learning → LLMs

Do journals keep pace with paradigm shifts — or does their published content drift away from their stated mission?

RQ 1

How well do TACL articles align with the journal's stated Aims & Scope?

RQ 2

Has this alignment changed from 2015 to 2025?

RQ 3

What are the main research topics, and how have their shares shifted?

Dataset: 620 TACL Articles via OpenAlex

620

TACL articles
2015 – 2025

OpenAlex

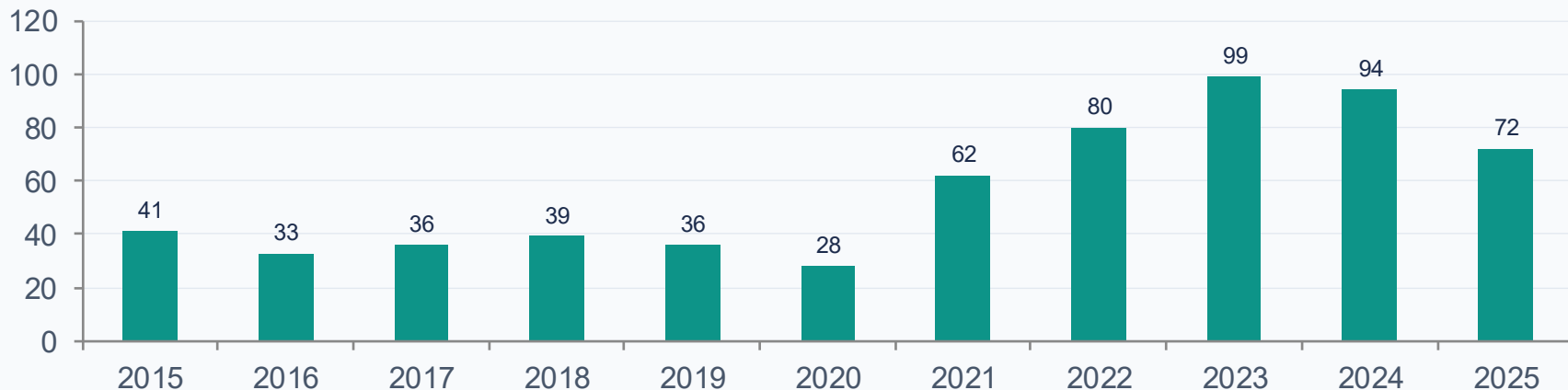
API source
DOI-verified

158

mean word count
per abstract

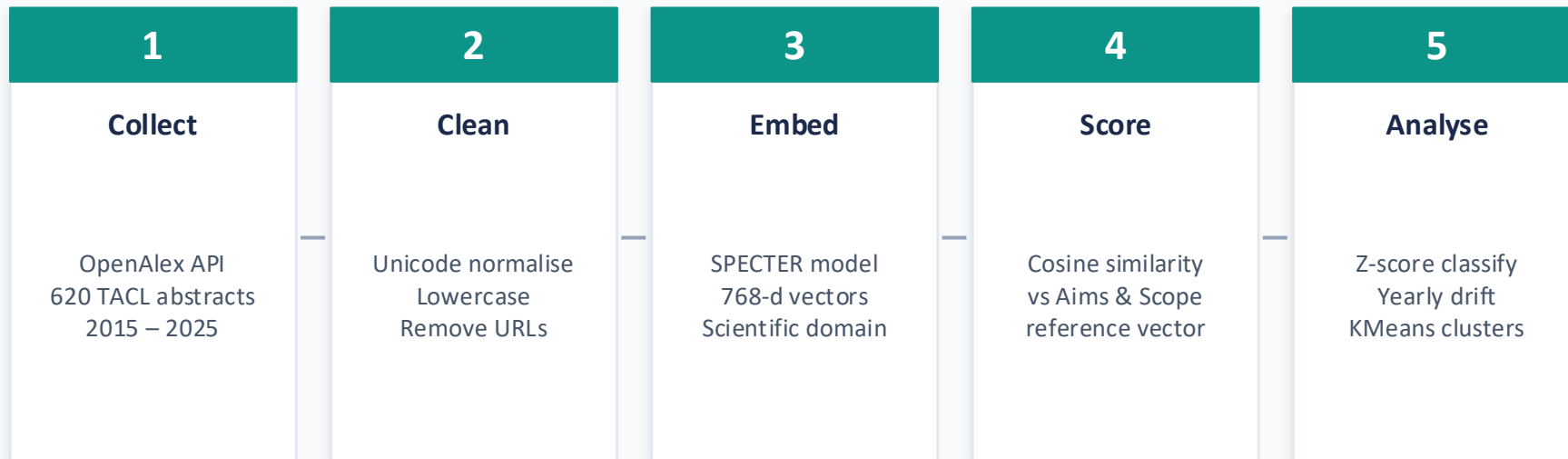
0 nulls

after cleaning
& deduplication



Note: 2025 is interpreted cautiously due to publication-date metadata granularity.

Pipeline: From Text to Alignment Score



SPECTER chosen over generic models: scientific vocabulary requires domain-specific pre-training · Aims & Scope embedded once as a fixed reference vector

Alignment Scoring & Classification

Cosine Similarity Formula

$$\text{align}(i) = (\mathbf{p}_i \cdot \mathbf{a}) / (\| \mathbf{p}_i \| \cdot \| \mathbf{a} \|)$$

Z-Score Classification

Category	Criterion	Count
High Alignment	$z > +1.0$	94 (15.2%)
Aligned	$-1.5 \leq z \leq +1.0$	476 (76.8%)
Outlier	$z < -1.5$	50 (8.1%)

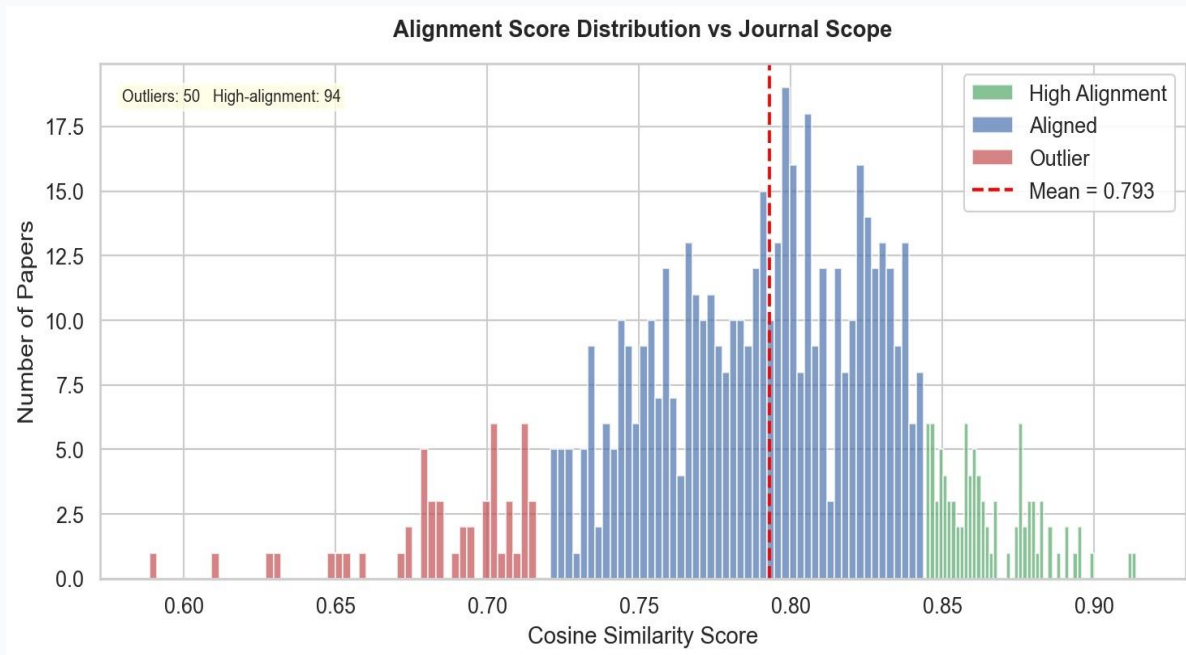
Key Metric

0.793

Mean Cosine Similarity
across all 620 TACL articles

*Most articles score 0.75 – 0.85,
indicating broad on-scope coverage*

Alignment Score Distribution



High Alignment

94 articles

15.2%

Aligned

476 articles

76.8%

Outliers

50 articles

8.1%

Outliers are not bad articles — they may be interdisciplinary or use terminology not reflected in the scope text.

Qualitative Inspection: Top & Bottom Articles

0.914

2024

Holmes: A Benchmark to Assess the Linguistic Competence of Language Models

0.589

2023

Removing Backdoors in Pre-trained Models by Regularized Continual Pre-training

0.911

2023

Efficient Methods for Natural Language Generation

0.609

2019

Learning End-to-End Goal-Oriented Dialog with Multiple Answers

0.899

2022

Quality at a Glance: An Audit of Web-Crawled Multilingual Datasets

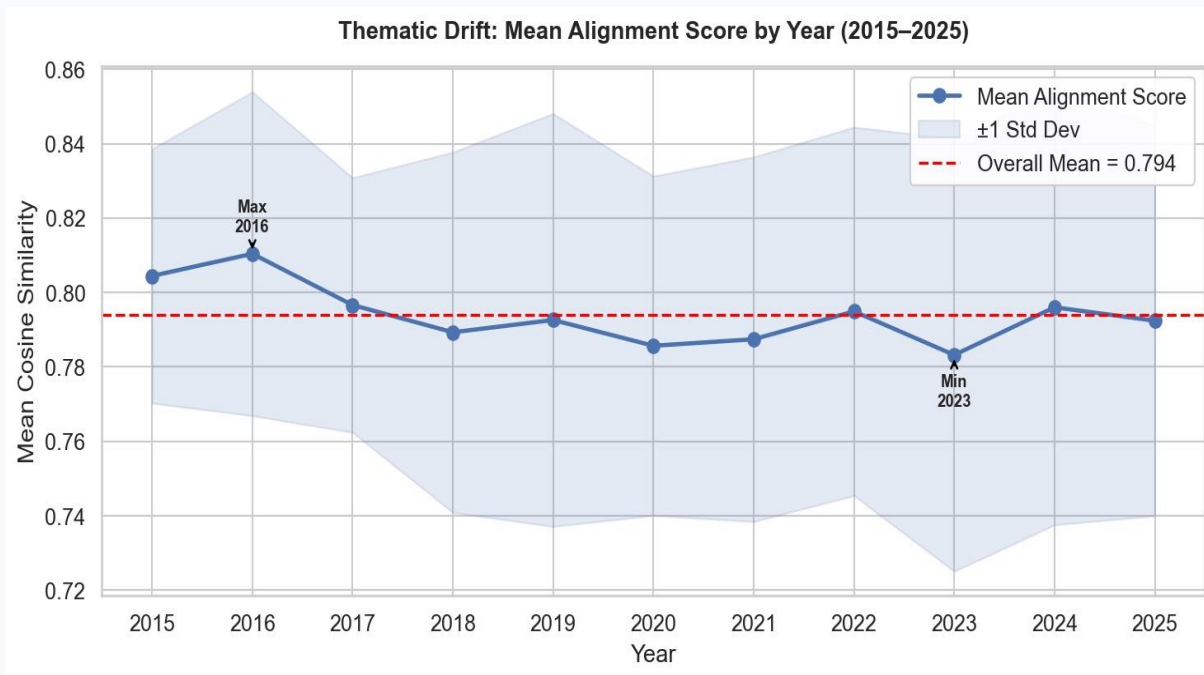
0.629

2021

Revisiting Few-shot Relation Classification

High-scoring articles address core NLP themes directly. Low-scoring articles are valid TACL contributions but framed around methods or applications underrepresented in the scope text.

Thematic Drift: Mean Alignment by Year



▲ Peak

2016

Highest mean alignment

▼ Dip

2023

Lowest mean alignment year

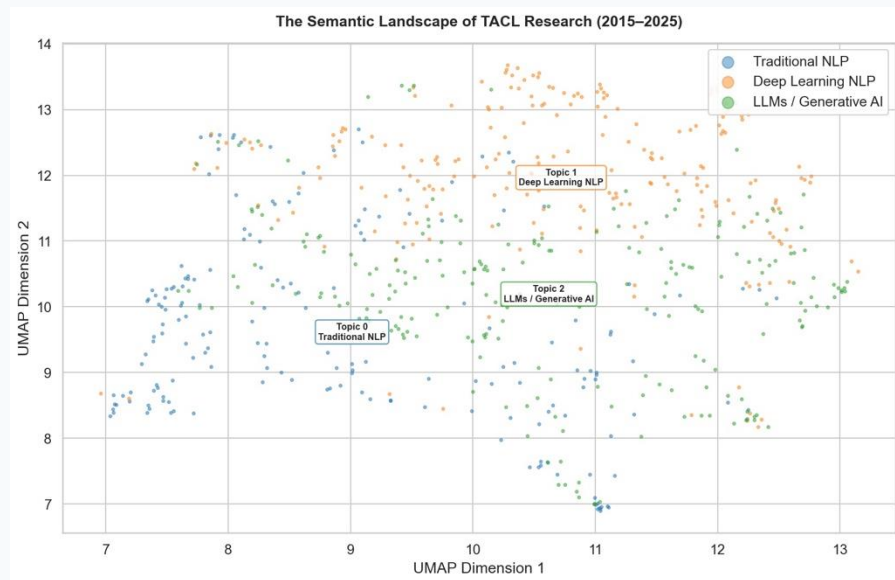
~ Range

0.028

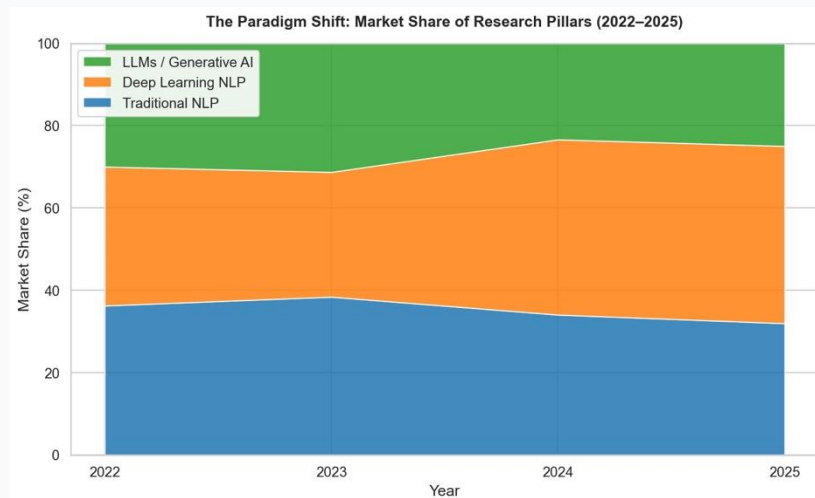
Max – Min spread (very small)

No strong thematic drift is observed. Year-to-year variation remains small. · 2025 is interpreted cautiously due to metadata granularity.

Research Topic Evolution (2022–2025)



Clusters assigned via *KMeans* ($k=3$, silhouette-optimised) on full 768-d space. UMAP used for 2D visualisation only.



Cluster	2022	2025	Trend
Traditional NLP	~36%	~32%	↓ slight
Deep Learning NLP	~34%	~43%	↑ growth
LLMs / Generative AI	~30%	~25%	↓ modest

Conclusion

RQ1

TACL is broadly on-scope. Mean alignment = 0.793. Only 8.1% of articles are outliers.

RQ2

Alignment is stable 2015–2025. The 0.028 range is too small to claim significant drift.

RQ3

Deep Learning NLP grew most (34→43%). LLMs / Generative AI remain visible but do not dominate the journal.